AMS Review (2025) 15:519–538
https://doi.org/10.1007/s13162-025-00323-y

THEORY/CONCEPTUAL

AI washing: A conceptual exploration

Samer Elhajjar[1] · **Omar S. Itani**[2]

Received: 26 November 2024 / Accepted: 8 September 2025 / Published online: 10 November 2025
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Abstract
This paper introduces and explores the concept of AI washing, a phenomenon where companies misrepresent or exaggerate their artificial intelligence (AI) capabilities to enhance marketing appeal and gain a competitive advantage. Despite its increasing prevalence, AI washing has received limited theoretical attention. Drawing on literature from similar practices, this paper develops a conceptual framework and typology to categorize various forms of AI washing. Theoretical implications include extending marketing ethics frameworks and existing theories to the domain of AI. The paper also highlights avenues for future empirical research, particularly in validating the proposed typology and exploring the consequences of AI washing on trust and brand reputation. From a practical standpoint, the paper offers recommendations for businesses to adopt more transparent AI marketing strategies and calls for regulatory interventions to mitigate the risks of AI washing. Finally, it discusses limitations and directions for further study.

Keywords AI washing · Business ethics · Technological ethics · Artificial intelligence · AI · Greenwashing · Bluewashing

Introduction

Artificial intelligence (AI) integration into products and services has rapidly gained prominence due to its potential to enhance efficiency, spur innovation, and improve decision-making across various industries (Mariani et al., 2023). However, the very ubiquity and promise of AI have also given rise to a critical, underexplored problem that this paper addresses: "AI washing," or the exaggeration and misrepresentation of AI capabilities to influence investors, consumers, and regulators. AI is widely regarded as a transformative force, reshaping markets, enhancing consumer experiences, and driving business growth (Kaggwa et al., 2024). Beyond commercial applications, AI is a key solution for addressing global challenges in healthcare, climate change, and transportation (Dwivedi et al., 2021). However, alongside the rise of AI, there is growing concern about the ethical implications of its adoption, particularly regarding transparency and accountability (Bankins & Formosa, 2023). A key issue emerging from this debate is AI washing—a phenomenon in which companies exaggerate or falsely claim AI capabilities in their products or services to gain a competitive edge, attract investors, or appeal to consumers. For example, a fitness tracker that claims to use AI to monitor sleep quality but only measures heart rate and movement while sleeping.

Building on established AI frameworks (e.g., Davenport & Ronanki, 2018; Huang & Rust, 2021), AI encompasses systems requiring human-like intelligence, such as learning, reasoning, adapting, and making autonomous decisions. Research suggests that many companies claiming to use AI do not deploy technologies that meet these criteria (Opera et al., 2024). This practice mirrors greenwashing (e.g., de Netto et al., 2020; Szabo & Webster, 2021), where companies falsely claim environmentally friendly practices, and bluewashing, where firms overstate their commitment to ethical and social responsibility (e.g., Pope & Wæraas, 2016; Delmas & Burbano, 2011). Like these deceptive branding strategies, AI washing distorts market dynamics, misleads consumers and users, and raises

✉ Samer Elhajjar
samer@nus.edu.sg

Omar S. Itani
omarsitani@gmail.com

[1] National University of Singapore Business School, Singapore, Singapore

[2] Adnan Kassar School of Business Lebanese American University, Beirut, Lebanon

serious ethical concerns. To address this, this study aims to examine the phenomenon of AI washing—the deceptive portrayal of AI capabilities—by creating a structured typology based on insights from corporate washing, marketing ethics, and technological misrepresentation. It aims to categorize various forms of AI washing, assess their ethical and strategic implications, and establish a conceptual framework for future studies.

The risks of AI washing extend beyond corporate misrepresentation, affecting both businesses and consumers. In a B2B context, AI washing is used to attract funding, secure partnerships, or gain technological legitimacy, potentially misleading investors and industry stakeholders. A case in point is Delphia, a financial technology firm fined $225,000 by the U.S. Securities and Exchange Commission (SEC) for making false and misleading claims about its AI capabilities. Delphia had asserted that its AI-driven models could predict market trends by analyzing data from retail clients. However, during an SEC investigation, the company admitted it had never developed such algorithms. Despite regulatory warnings, Delphia continued making deceptive AI claims in its marketing materials. This case exemplifies a broader concern: AI washing is an ethical issue and a systemic problem with market-wide consequences, affecting corporate valuations, regulatory enforcement, and public perception of AI technologies.

AI washing creates misplaced trust in AI-powered products. For example, some tech companies have marketed AI-driven personalization, automation, and security features that, upon closer inspection, rely on basic rule-based algorithms or heavy human intervention rather than true AI functionalities (Kumar et al., 2024). For example, Amazon's Just Walk Out technology, initially promoted as an AI-driven solution for automated retail checkout, came under scrutiny when investigations revealed it relied heavily on human labor for transaction review and data labeling, particularly with over 1,000 workers involved. While Amazon initially denied these reports, stating that human involvement was limited to labeling and annotating data to improve machine learning models and not live transaction processing, further reporting confirmed that substantial manual oversight was required for the technology to function at scale. Eventually, Amazon decided to discontinue the Just Walk Out technology.

Despite these growing concerns, AI washing remains an underexplored phenomenon in the realms of marketing ethics and corporate responsibility. While early discussions on AI governance have focused on fairness, accountability, and bias (Mittelstadt et al., 2016; Birkstedt et al., 2023), there is a lack of research on how AI misrepresentation specifically affects marketing and consumer trust. This paper addresses this gap by offering a conceptual exploration of AI washing, identifying its drivers, consequences, and implications for businesses, consumers, and regulators.

We adopted a conceptual research approach synthesizing insights from corporate washing practices, marketing ethics, and technological misrepresentation. By examining AI washing within a broader framework of deceptive branding, this paper develops a structured typology that categorizes different forms of AI misrepresentation and outlines their ethical and strategic implications. This approach facilitates a more systematic understanding of AI washing and lays the groundwork for future empirical studies. To guide this exploration, this study addresses the following research questions:

RQ1 What are the core forms of AI washing in marketing, and how can they be conceptually distinguished through a marketing ethics lens?

RQ2 What are the key drivers and market consequences of AI washing?

RQ3 How can businesses, regulators, and consumers mitigate the risks associated with AI washing?

This study makes three key contributions. First, it defines AI washing and develops a structured typology to classify AI washing, differentiating between its manifestations in B2B and consumer markets. Second, it examines the ethical risks and market implications of AI washing, providing a theoretical framework for understanding its broader effects. Third, it proposes practical strategies for mitigating AI washing, helping businesses and regulators promote transparency and responsible AI marketing practices. It enables consumers to understand the problem and avoid being misled, deceived, or trapped by AI washing. To date, no studies have examined the three research questions raised, and to our knowledge, very few researchers have investigated AI washing as a noteworthy phenomenon (e.g., Al Haddi, 2024; Leffrang & Mueller, 2023).

As AI continues to reshape industries and societies, understanding the challenges posed by AI washing is crucial. This paper contributes to the broader conversation on responsible AI adoption by addressing these challenges, fostering a more transparent and accountable AI-driven future. The first step in our research approach involved conducting an extensive literature review of the washing practices in various fields to identify parallels with AI washing. By examining the core characteristics of these practices, we were able to map the similarities and differences in misrepresentation across industries and subjects. This theoretical investigation was essential for grounding our typology of AI washing in well-established ethical and marketing frameworks. This typology is designed to serve as a foundation for future empirical testing and offers a structured lens through which

AI marketing claims can be analyzed. We also used this literature review to define the key dimensions of AI washing and identify forms of misrepresentation specific to the marketing of AI technologies.

Literature review

AI and its role in business and society

AI has become one of the most transformative technologies in recent decades, influencing various sectors such as healthcare, finance, manufacturing, and retail (Dwivedi et al., 2021). AI refers to the use of advanced algorithms, machine learning, and data-driven processes to mimic human cognitive functions such as decision-making, problem-solving, and pattern recognition (Brynjolfsson & McAfee, 2017; Huang & Rust, 2021). The rapid adoption of AI has been driven by its ability to enhance operational efficiency, improve decision-making accuracy, and personalize customer experiences (Dwivedi et al., 2021). For example, AI-driven recommendation engines are widely used by e-commerce companies to personalize shopping experiences (Acharya et al., 2023), while AI algorithms in healthcare have been designed to assist doctors in diagnosing diseases more accurately (Davenport & Ronanki, 2018; Khanna et al., 2022).

Despite these advancements, there is a growing concern about how companies present and communicate their AI capabilities to stakeholders (e.g., Du & Xie, 2021; Zhan et al., 2024). As AI becomes a buzzword representing cutting-edge innovation, businesses face increasing pressure to align with this technological trend. However, not all companies that claim to be AI-driven truly leverage AI systems (Akter et al., 2021; Satornino et al., 2024). AI washing refers to the discrepancy between a company's actual technological capabilities and how it is marketed to the public. This trend aligns with a broader framework of technological hype, where new technologies are overpromoted to create inflated expectations that may not be realized in practice (Brennen et al., 2020).

Technological hype and its connection to AI washing

Technological hype is a well-established phenomenon that describes the cycle of overpromising and underdelivering that often accompanies new technological innovations (Van Lente et al., 2013). As companies and investors become excited about emerging technologies, the potential of these technologies is often overstated, leading to unrealistic expectations. The Gartner Hype Cycle offers a model for understanding this process (Dedehayir & Steinert, 2016). It outlines five stages that technologies typically go through: the "technology trigger," the "peak of inflated expectations," the "trough of disillusionment," the "slope of enlightenment," and finally, the "plateau of productivity" (Fenn & Raskino, 2008).

AI washing, like other instances of technological hype, exploits the early stages of this cycle, particularly the "peak of inflated expectations." In this phase, companies use the media's portrayal of AI as a transformative tool that can solve complex industry problems. This creates a situation where companies feel compelled to present themselves as leaders in AI innovation to remain competitive and relevant. However, in many cases, these companies often lack the sophisticated AI capabilities they claim to possess. This practice of overstating AI's role in products and services has detrimental consequences, as it sets the stage for disillusionment among consumers, buyers, investors, and the public when the real capabilities of AI are revealed to be more limited than initially claimed (e.g., Pasquale, 2020a, b; LaGrandeur, 2024).

Technological hype is not limited to AI washing; however, parallels can be drawn to other forms of misrepresentation in business practices, such as greenwashing and bluewashing (e.g., de Netto et al., 2020), where companies misrepresent their environmental or social responsibility initiatives, respectively. Both practices highlight a pattern of leveraging perceived societal values and regulatory requirements in sustainability, ethics, or corporate responsibility for branding and market positioning rather than for genuine change. AI washing follows a similar trajectory in which firms exploit the growing appeal of AI by promoting products as artificially intelligent, even when they do not incorporate any AI technologies (Kumar et al., 2024). Understanding these deceptive strategies provides a foundation for categorizing AI washing within a broader framework of corporate misrepresentation, helping to define its typology and implications.

Greenwashing: Definition, characteristics, drivers, and consequences

Greenwashing refers to the practice of misleading stakeholders about the environmental benefits of a company's products, services, or practices (Torelli et al., 2020). This term originated in the 1980 s, initially describing how companies promoted superficial environmental initiatives to create a false impression of eco-friendliness (Polonsky et al., 2010). Typically, greenwashing involves companies making exaggerated, vague, or unsubstantiated claims about their sustainability efforts, such as using buzzwords like

"eco-friendly," "green," or "sustainably sourced" without providing concrete evidence to support these claims (Szabo & Webster, 2021). The primary objective of greenwashing is to capitalize on the growing demand for environmentally responsible products and services, even when the company's actual environmental impact remains unchanged or negligible (Du, 2015).

The characteristics of greenwashing are often subtle yet manipulative (Torelli et al., 2020). Companies frequently highlight small, token environmental initiatives, such as using recycled materials in product packaging or reducing water consumption in one segment of their operations, while continuing to engage in environmentally harmful practices in other areas (Lyon et al., 2015a, b). For instance, a fast-fashion brand may promote a new line of eco-friendly clothing made from organic cotton while ignoring the broader environmental and social impacts of its supply chain, including poor labor practices and high levels of textile waste.

Several key factors drive companies to engage in greenwashing. One of the most significant drivers is competitive pressure. In industries where sustainability is increasingly seen as a competitive advantage, companies must highlight their environmental credentials to differentiate themselves from competitors (Parguel et al., 2011). Another major driver is investor expectations, particularly as more investors prioritize Environmental, Social, and Governance (ESG) criteria in their decision-making processes. Companies may exaggerate their green initiatives to attract investment or maintain favor with ESG-conscious investors (Lyon & Montgomery, 2015a, b). Furthermore, consumer demand plays a central role in driving greenwashing. As consumers become more aware of environmental issues and more likely to choose eco-friendly products, companies seek to align with these values, even if their sustainability efforts are limited (Leonidou & Skarmeas, 2017a, b). Finally, a lack of regulatory oversight allows greenwashing to proliferate, as companies face minimal consequences for making vague or exaggerated environmental claims (Testa et al., 2018).

The consequences of greenwashing are significant and far-reaching. One of the most immediate effects is the loss of consumer trust. When consumers discover that a company has engaged in greenwashing, they are less likely to trust the brand and its future environmental claims, leading to reduced brand loyalty and potentially lower sales (Nyilasy et al., 2014a, b). Moreover, greenwashing can cause reputational damage, which has long-term implications for customer retention, investor confidence, and the company's public image (Torelli et al., 2019). As greenwashing becomes more prevalent, it also draws regulatory scrutiny, with governments and regulatory bodies imposing stricter guidelines for environmental marketing and product claims. This can result in fines, lawsuits, and the need for corrective advertising or more transparent reporting (Lyon & Montgomery, 2015a, b).

Bluewashing: Definition, characteristics, drivers, and consequences

Bluewashing, like greenwashing, involves exaggerating or fabricating a company's ethical and socially responsible practices, but it focuses specifically on corporate social responsibility (CSR) (Pope & Wæraas, 2016). The term emerged from companies that aligned themselves with international organizations, such as the United Nations (often symbolized by the blue UN logo), to falsely signal their commitment to social and ethical standards, even when their business practices fell short of these standards (Utting, 2007). Bluewashing typically involves companies making symbolic gestures, such as supporting charitable causes or claiming to uphold high labor standards, without taking any meaningful action to improve their actual impact on social issues (Laufer, 2003; Pope & Wæraas, 2016).

The characteristics of bluewashing are often challenging to detect because they rely on the strategic use of vague and symbolic actions. For example, a company might emphasize its contributions to a charitable cause while continuing to engage in poor labor practices or exploitative supply chains. Like greenwashing, bluewashing is often characterized by empty language that appeals to consumers' and investors' values without offering verifiable proof of ethical behavior (Berliner & Prakash, 2015). By making superficial changes or selectively highlighting certain aspects of their operations, companies can create a misleading image of social responsibility that conceals their broader negative impact (Berliner & Prakash, 2015).

The drivers of bluewashing are similar to those of greenwashing and can be traced to several key factors. Social pressure is a primary driver, as companies must demonstrate their commitment to CSR in response to growing societal concerns about human rights, labor conditions, and diversity (Scherer et al., 2013). In addition, public relations strategy plays a significant role, with companies using bluewashing to bolster their public image and enhance brand loyalty by associating themselves with socially desirable causes (Lubin & Esty, 2010). Investor demand for companies prioritizing ethical business practices further incentivizes bluewashing, particularly as more institutional investors adopt ESG criteria in their portfolio decisions (Pérez & Goh, 2021a, b). Finally, as with greenwashing, the lack of stringent regulations governing CSR reporting and ethical claims allows companies to engage in bluewashing without significant consequences (Utting, 2007).

The consequences of bluewashing are similar to those of greenwashing, particularly in terms of consumer trust and reputational damage (Ruiz-Blanco et al., 2022). When companies are exposed to bluewashing, they risk losing the trust of consumers and investors, particularly those who prioritize ethical business practices. This can lead to reduced sales, investor divestment, and negative media attention (Leonidou & Skarmeas, 2017a, b). Moreover, bluewashing contributes to the overall devaluation of genuine CSR efforts, as it fosters skepticism among consumers and investors about the sincerity of all CSR claims. Over time, this dilutes the impact of real CSR initiatives and makes it harder for genuinely responsible companies to distinguish themselves (Scherer et al., 2013). Also, as bluewashing becomes more prevalent, firms that engage in these practices may face increased regulatory scrutiny, particularly as governments and NGOs push for greater transparency in CSR efforts (Gatti et al., 2019).

Table 1 summarizes the key drivers behind washing practices, including competitive pressures, investor expectations, and lack of regulatory oversight. In contrast, Table 2 outlines the potential consequences of these practices, such as loss of consumer trust, reputational damage, and hindered innovation adoption.

Table 1 Summary of drivers of washing practices

Driver	Summary	References
Competitive Pressures	Companies use washing practices to differentiate themselves from competitors in industries where sustainability or innovation is highly valued.	Parguel et al. (2011); Pérez and Goh (2021a, b)
Investor Expectations	Firms overstate their environmental or technological capabilities to meet investor demand for ESG and innovation-driven business models.	Lyon & Montgomery, 2015a, b; Pasquale, (2020a, b)
Consumer Demand	Consumer preferences for ethical, sustainable, or innovative products encourage companies to exaggerate their commitments in these areas.	Bowen and Aragon-Correa (2014); Leonidou and Skarmeas (2017a, b)
Lack of Regulatory Oversight	Weak regulatory frameworks allow companies to engage in washing practices without significant penalties.	Floridi et al. (2018a, b; Testa et al. (2018)
Social and Political Pressure	Social expectations surrounding sustainability and ethical behavior can lead companies to make exaggerated claims in an effort to align with these values.	Testa et al. (2018); Floridi et al. (2018a, b)
Public Relations Strategy	Washing practices are often driven by a desire to enhance brand image and align with positive values such as innovation, sustainability, or ethics.	Lubin and Esty (2010); Torelli et al. (2019)

Table 2 Summary of consequences of washing practices

Consequence	Summary	Reference
Loss of Consumer Trust	Once consumers discover deceptive practices, they are less likely to trust the brand, which in turn reduces loyalty and market share. Greenwashing specifically diminishes trust in sustainability efforts.	Leonidou & Skarmeas, 2017a, b; Nyilasy et al., 2014a, b; Szabo and Webster, (2020); Guo et al. (2015); Chen and Chang (2012); Rana and Dwivedi (2020)
Reputational Damage	Misleading stakeholders results in long-term reputational harm, reducing consumer goodwill, investor confidence, and the ability to attract talent. Greenwashing accelerates reputational damage, particularly when negative experiences spread rapidly through social media.	Torelli et al. (2019); Pérez and Goh (2021a, b; Macotela, (2016); Nucci and Hibberd (2021)
Increased Regulatory Scrutiny	Washing practices often lead to stricter regulatory standards, forcing companies to improve their reporting or face legal penalties.	Lyon & Montgomery, 2015a, b; Floridi et al. (2018a, b)
Market Instability	Exaggerated claims can contribute to speculative bubbles, leading to financial losses for investors and market instability.	Pérez and Goh (2021a, b)
Hindered Adoption of Innovation	Misleading claims about technological or environmental capabilities can create scepticism, slowing the adoption of genuine innovations.	Pasquale, 2020a, b; Mittelstadt, (2019)
Reduction of Consumer Engagement	Consumers tend to engage less with brands perceived as guilty of greenwashing, resulting in decreased social media interaction and overall engagement, which weakens the brand-consumer relationship.	Qayyum et al. (2022)
Weakening of Brand Attachment	Greenwashing diminishes consumers' emotional attachment to a brand, as deceptive actions provoke feelings of betrayal and disappointment, which negatively affect the brand's image and purchase intent.	Grégoire and Fisher (2007); Shi &Sun (2024); Lopes and Veiga (2019); Wood (2015)
Risks of Brand Boycott	Perceptions of greenwashing increase the likelihood of consumer boycotts, especially among those with strong ethical values.	Paassilta (2021); Xiao et al. (2022)
Sales and Customer Acquisition	Negative word-of-mouth fuelled by consumer experiences with greenwashing can reduce sales and hinder new customer acquisition.	Guerreiro and Pacheco (2021); Martin and Lueg (2013)

Theoretical framework: Understanding AI washing

The theoretical framework for understanding AI washing draws on multiple theories that explain why companies might engage in this deceptive practice and its broader implications for stakeholder relationships, competitive positioning, and market dynamics. As AI technologies continue to generate excitement and drive innovation across industries, the pressure on companies to align with these technological trends has intensified. There is growing concern that some firms may exaggerate their AI capabilities or misrepresent the functionality of AI-driven products, contributing to AI washing. (Opera et al., 2024). This framework builds on signaling theory, legitimacy theory, stakeholder theory, and ethics washing theory to provide a comprehensive explanation of the motivations behind AI washing and its consequences.

Signaling theory

Signaling theory offers a crucial lens through which to understand AI washing, as it explains how companies use signals to convey information to external stakeholders, particularly in contexts where information asymmetry exists (Connelly et al., 2011). AI washing is essentially a form of signaling, where firms attempt to signal (Connelly et al., 2011) that they are technologically advanced and innovating through the use of AI, regardless of whether these claims are substantiated. In markets where AI is highly valued, companies employ AI washing to signal their alignment with technological trends, thereby appealing to investors, customers, and competitors (Spence, 1973).

The underlying principle of signaling theory is that companies with superior capabilities send signals to distinguish themselves from competitors (Connelly et al., 2024). However, companies without those capabilities may attempt to mimic these signals to gain similar advantages (Kirmani & Rao, 2000). AI washing is an example of this mimetic behavior, as companies that lack true AI capabilities market themselves as AI-driven, hoping to gain investment or consumer favor. The issue of signal credibility becomes central to AI washing. When companies engage in AI washing and are later exposed, the credibility of their signals is damaged, leading to erosion of trust and potential long-term reputational harm (Connelly et al., 2011).

Applying signaling theory to AI washing also helps explain why companies engage in this practice despite the risks involved. The perception of being AI-driven offers immediate benefits, such as attracting investors eager to fund innovative AI projects, winning over consumers interested in cutting-edge technology, or gaining media attention for perceived leadership in AI (Akter et al., 2023). In short, AI washing is driven by the need to appear innovative in a rapidly evolving marketplace where true AI capabilities can be complex for outsiders to verify.

Legitimacy theory

Legitimacy theory is another key theoretical perspective that helps explain the prevalence of AI washing. This theory suggests that companies seek to align themselves with society's norms, values, and expectations to maintain their legitimacy in the eyes of stakeholders (Suchman, 1995). In today's business environment, AI is increasingly seen as a marker of innovation, efficiency, and future-readiness (Elhajjar, 2024). Companies that fail to demonstrate their engagement with AI risk being perceived as lagging behind competitors or being technologically outdated.

AI washing can be understood as a response to the legitimacy gap—the discrepancy between what society or stakeholders expect from a company and what the company can deliver (Deegan, 2002). In industries where AI is expected to drive the future of innovation, companies that cannot implement advanced AI systems may resort to AI washing to maintain legitimacy. This is especially true in sectors such as finance and healthcare, where AI is viewed as crucial for enhancing decision-making, optimizing processes, and personalizing consumer experiences (Dwivedi et al., 2021). Even when claiming to be AI-driven, companies attempt to close the legitimacy gap and remain relevant in a competitive marketplace, even when this is not entirely true.

While AI washing may temporarily bolster a company's legitimacy, Legitimacy theory emphasizes the long-term risks of deceptive practices (Thomas & Lamm, 2012). Suppose stakeholders discover that a company has misrepresented its AI capabilities. In that case, the resulting legitimacy crisis can lead to significant reputational damage, loss of consumer and investor trust, and potential regulatory consequences (Thomas & Lamm, 2012). Therefore, while AI washing may help companies appear legitimate in the short term, it can ultimately undermine their standing in the marketplace if the deception is uncovered.

Stakeholder theory

Stakeholder theory, introduced by Freeman (1984), focuses on how businesses interact with various stakeholder groups, including customers, employees, investors, regulators, and the broader community. According to this theory, companies must balance the interests of their stakeholders to ensure long-term success and maintain trust (Friedman & Miles, 2002). AI washing can be understood as a tactic companies use to align with the expectations of

key stakeholder groups, particularly investors and consumers, who increasingly prioritize AI innovation in their decision-making processes.

From the perspective of stakeholder theory, companies engage in AI washing to satisfy the expectations of stakeholders who view AI as a key differentiator in the marketplace. For investors, AI is seen as a critical area of innovation that promises future growth and high returns, making companies that claim to be AI-driven more attractive investment opportunities (Pasquale, 2020a, b). Similarly, consumers who seek out products and services powered by AI may be drawn to companies that market themselves as leveraging advanced AI technologies, even if the reality falls short of these claims.

However, stakeholder theory also highlights the potential backlash from stakeholders when they discover that they have been misled. Companies that engage in AI washing risk losing the trust of consumers, investors, and other stakeholders when the discrepancy between their claims and actual practices becomes evident. This loss of trust can have far-reaching consequences, including customer churn, investor divestment, and negative media attention. Therefore, while AI washing may provide short-term benefits by aligning with stakeholder expectations, it ultimately undermines the company's relationship with its stakeholders and jeopardizes long-term success.

Within the stakeholder theory framework, ethical responsibility extends beyond profit generation to encompass the duty of businesses to act transparently and fairly in their interactions with all stakeholders (Heath & Norman, 2004). Corporate communication, particularly regarding technological capabilities, must align with ethical standards to maintain legitimacy and trust (Freeman, 1984). AI washing—misrepresenting the capabilities of AI—undermines these ethical obligations and carries significant consequences (Peifer & Newman, 2020).

The erosion of trust is among the most immediate effects of AI washing. Stakeholders, including consumers, employees, and investors, depend on accurate disclosures to make informed decisions. When firms exaggerate AI capabilities, they create a false perception of technological progress, leading to disillusionment when the technology fails to meet expectations. Studies on corporate deception indicate that once trust is compromised, recovery is challenging and may result in permanent reputational damage (Kim et al., 2009; Chiou & Lee, 2023).

Regulatory backlash is an inevitable outcome as AI governance frameworks evolve. Governments and regulatory bodies worldwide are tightening oversight on AI-related claims to prevent misleading practices (Smuha, 2021). Misleading AI disclosures could result in legal penalties, litigation, and increased scrutiny, as seen in other cases of deceptive marketing practices in the technology sector (Pasquale, 2023).

Beyond immediate legal risks, AI washing fosters market disillusionment, negatively impacting innovation. When multiple firms engage in exaggerated AI claims, stakeholders become skeptical of genuine advancements in AI technology. Historical parallels can be drawn to periods of excessive technological hype, such as the dot-com bubble, where exaggerated capabilities led to widespread skepticism and hesitation in investment (Cassidy, 2002). In this context, AI washing can erode investor confidence and hinder adoption rates, ultimately impacting even companies operating ethically.

Innovation stagnation is another consequence, as firms divert resources toward marketing narratives rather than substantive AI development. Overstatement of AI capabilities may lead to misallocated funding, where investors and organizations prioritize firms with inflated claims rather than those with genuine advancements (Brynjolfsson & McAfee, 2017; Thais, 2024). When expectations are misaligned with actual technological feasibility, businesses face setbacks, and the overall progress of AI-driven industries is hampered.

Reputational damage further complicates the issue (Peifer & Newman, 2020). The credibility of their claims can shape public perception of AI-driven companies, and once AI washing is exposed, the damage extends beyond the immediate controversy (Peifer & Newman, 2020). Companies caught engaging in AI washing may struggle to regain public trust (Peifer & Newman, 2020), as reputational risks can lead to reduced customer loyalty, weakened investor confidence, and a diminished competitive positioning.

Through the lens of stakeholder theory, AI washing represents a failure of ethical corporate governance (Heath & Norman, 2004). Pursuing short-term gains through misleading AI claims ultimately destabilizes long-term stakeholder relationships, leading to regulatory intervention, skepticism, and a diminished market for genuine AI innovation.

Ethics washing theory

Ethics washing involves superficially signaling a commitment to ethical standards or values without truly intending to implement or uphold them (e.g., Van Maanen, 2022; Yeung et al., 2020). It is similar to ethics shopping, where no clear ethical lists or standards are circulated, causing confusion and inconsistency, as seen in the case of AI use by some organizations today (e.g., Van Maanen, 2022; Wagner, 2018). Ethics washing reflects the vulnerability in applying ethical standards in ways that benefit organizations, making it more challenging to control new phenomena. It begins with the acknowledgment that limited ethical lists and guidelines

exist, leading to confusion, information overload, and the practice of ethics shopping, which is linked to ethics washing. Ethical washing can impact decision-making processes within organizations, as it mainly involves adopting general ethical codes and forming symbolic ethics committees. It aims to create the appearance of ethical responsibility in organizational decision-making while engaging in questionable practices. This phenomenon contradicts the concept of Corporate Digital Responsibility (CDR) and organizational ethical decision-making in marketing (e.g., Marcoux, 2025). While the literature on ethics washing offers an important perspective on AI washing, considering the positive theory of marketing (Hunt & Vitell, 1986) could help connect the philosophical aspects of ethics washing to practical ethical decision-making.

AI washing is a subtype of ethics washing, involving the exaggeration or fabrication of claims about artificial intelligence to enhance perception or secure market leverage, which raises considerable ethical concerns. A related theory is ethics washing, reflecting deceptive virtue signaling or performative ethics (Schultz & Seele, 2023; Seele & Schultz, 2022). Ethics washing involves pretending to consider ethics in order to enhance an organization's reputation without a genuine commitment to ethical principles or meaningful change. This resembles AI washing or machinewashing in the context of business ethics (Seele & Schultz, 2022; Schultz et al., 2024). Companies engaging in ethics washing use misleading language, visuals, omissions, symbolic gestures, and lobbying, contributing to deceptive virtue signaling (Seele & Schultz, 2022). Critics have labeled these voluntary codes as 'ethics washing,' particularly given compelling evidence that the tech industry often fails to meet its self-imposed obligations (Yeung et al., 2020).

In the realm of AI, ethics washing, like AI washing, involves companies making vague or misleading claims about the ethical utilization or capabilities of their AI systems. This often serves as a public relations strategy to address criticism without implementing meaningful ethical reforms (e.g., Attard-Frost et al., 2023; Morley et al., 2021; Van Maanen, 2022). Current AI ethics guidelines overly focus on algorithmic decision-making, neglecting essential components such as fairness, accountability, sustainability, and transparency in AI-related business dealings (Attard-Frost et al., 2023). Competitive pressures, ethical facades, corporate secrecy, and various harmful practices significantly undermine these crucial aspects (Attard-Frost et al., 2023).

Ferrell et al. (2024) modernized Hunt and Vitell's well-known positive theory of marketing ethics, originally developed in 1986, to fit the context of AI. Their new model emphasizes individual decision-making and reflects the decision process used by autonomous AI systems. According to this updated framework, ethical considerations in AI marketing are shaped by industry, cultural, organizational, and legal principles (Ferrell et al., 2024). These principles influence the risks and outcomes associated with AI algorithms, guiding AI actions toward ethical results. Principles such as transparency, accountability, and fairness should be turned into concrete rules that direct the development of AI algorithms (Ferrell & Ferrell, 2024). This translation process should be part of the organization's broader strategic and compliance efforts. Instead of viewing AI solely as a tool for boosting efficiency, organizations should see it as a way to responsibly manage both risks and opportunities, contributing to a better society (Ferrell & Ferrell, 2024). Human oversight remains crucial to ensure AI is properly managed and cannot bypass human control.

The positive theory of marketing ethics combines both deontological and teleological ethical approaches, emphasizing not only specific behaviors but also their consequences (Hunt & Vitell, 1986). This perspective is crucial when developing an ethical framework for AI transparency, taking into account organizational values, marketing strategies, and consumer decision-making processes. Ultimately, ethical AI decision-making should integrate philosophical principles with practical factors, especially those related to organizational decision-making and consumer choices. According to the positive theory of marketing ethics (e.g., Ferrell et al., 2024), the lack of clear principles—such as cultural, industry, organizational, and legal—causes many organizations to engage in AI washing, leading to dilemmas, issues, and biases related to AI. Consequently, this can result in vulnerable algorithmic rules that cannot ethically manage organizational AI actions and practices, thereby raising ethical concerns and leading to problematic outcomes.

Leslie (2019) highlights that four core AI ethics principles must prevail: fairness, accountability, sustainability, and transparency. Leslie (2019) emphasizes that AI developers and users should strive to prevent harm from biased or discriminatory outcomes and actively challenge unfair treatment. The responsibility for justifying AI-driven decisions lies with human creators and operators, who must be held accountable through effective mechanisms. Moreover, AI should be harnessed to yield positive, transformative, and sustainable benefits for individuals and society, promoting long-term social and environmental well-being. Transparency is essential, mandating that developers and users comprehend the rationale behind specific outcomes in defined contexts, thereby making decision processes understandable to all stakeholders.

Attard-Frost et al. (2023) extend Leslie's exploration by scrutinizing these four central AI ethics principles through a business perspective. They conclude that AI practices should be designed to prevent harm from biased results,

ensuring fairness for all individuals and groups. Accountability requires human creators and operators within AI-driven enterprises to justify their actions, supported by robust accountability mechanisms. Sustainable AI initiatives should aim to deliver lasting benefits to individuals and society. Transparency remains vital, as it requires the public or governmental bodies to understand how and why AI models work in specific contexts, thereby clarifying the rationale behind decisions.

To effectively counter ethics washing, it is crucial not to view ethical language merely as a tool to minimize, but to differentiate between superficial, performative ethics and authentic commitments to ethical principles that foster deeper self-reflection and comprehension (Bietti, 2020). While philosophers may need to adapt their strategies to remain relevant amid the swiftly evolving and politically charged technological landscape, the profound insights and nuanced perspectives offered by philosophy are essential to ongoing discussions, especially concerning technology governance (Bietti, 2020).

AI washing: Definition and conceptual framework

Definition of AI washing

Building on insights from the literature review and the theoretical frameworks used, we define AI washing. AI washing is the practice where companies deliberately overstate, exaggerate, or misrepresent their use or integration of AI in their products, services, or operations. This misrepresentation aims to boost the company's competitive edge, brand image, or investment appeal by leveraging the increasing societal and market demand for AI-driven innovations, even when the underlying technology lacks real AI sophistication. AI washing can show up through superficial claims about AI, strategic exaggerations aimed at investors, symbolic links to AI to boost public perception, or compliance-driven statements made to avoid regulatory scrutiny. As with those washing forms, AI washing exploits the information asymmetry between companies and stakeholders, including consumers, investors, regulators, and industry peers, who may lack the expertise or resources to verify the company's AI claims.

Conceptual framework for AI washing

The AI washing conceptual framework (Fig. 1) is informed by signaling, legitimacy, and stakeholder management theories, which explain the drivers, mechanisms, and consequences of this phenomenon. Below is the proposed conceptual framework that integrates these elements. The proposed framework establishes the key drivers, underlying mechanisms, and consequential outcomes of AI washing. At the same time, the accompanying typology classifies the various ways in which firms engage in deceptive AI-related claims. The interconnections among these elements present a rich foundation for future research, facilitating a better understanding of AI washing's implications for businesses, consumers, and regulatory bodies. For instance, subsequent studies could evaluate the efficacy of regulatory interventions and transparency initiatives in mitigating AI washing, assess how different industries respond to AI-related disclosure mandates, and investigate whether more stringent regulations effectively curtail misleading AI claims. Moreover, further inquiry into consumer behavior is crucial, examining how individuals detect AI washing, its influence on consumer trust, and the role of media literacy in safeguarding against deceptive AI marketing. Building on this framework, scholars can advance theoretical models that trace the evolution of AI washing and propose actionable recommendations for organizations and policymakers to promote the adoption of ethical AI and responsible marketing practices (See Table 3).

Drivers of AI washing

The framework initiates by identifying the key drivers that influence AI washing. These drivers are systematically categorized into two primary groups based on their relationship to the firm: external drivers, which encompass market forces and environmental factors outside the organization, and internal drivers, which include resources, capabilities, and internal processes within the firm.

External drivers

One primary external driver is market pressure. As artificial intelligence becomes increasingly integrated into various industries, companies often feel the need to appear aligned with AI trends, even if they lack actual AI functionality. Investor expectations amplify this pressure. Since AI is widely viewed as a key area of innovation and future growth, companies are motivated to claim AI adoption to attract funding and boost their valuation, especially in technology-driven sectors. Another significant external driver is the influence of media and public perception. The media often portrays AI as a game-changing force, leading companies to present themselves as leaders in AI in order to maintain relevance and competitiveness in the eyes of consumers and stakeholders. Additionally, evolving regulatory landscapes play a role. In industries where the use of ethical and

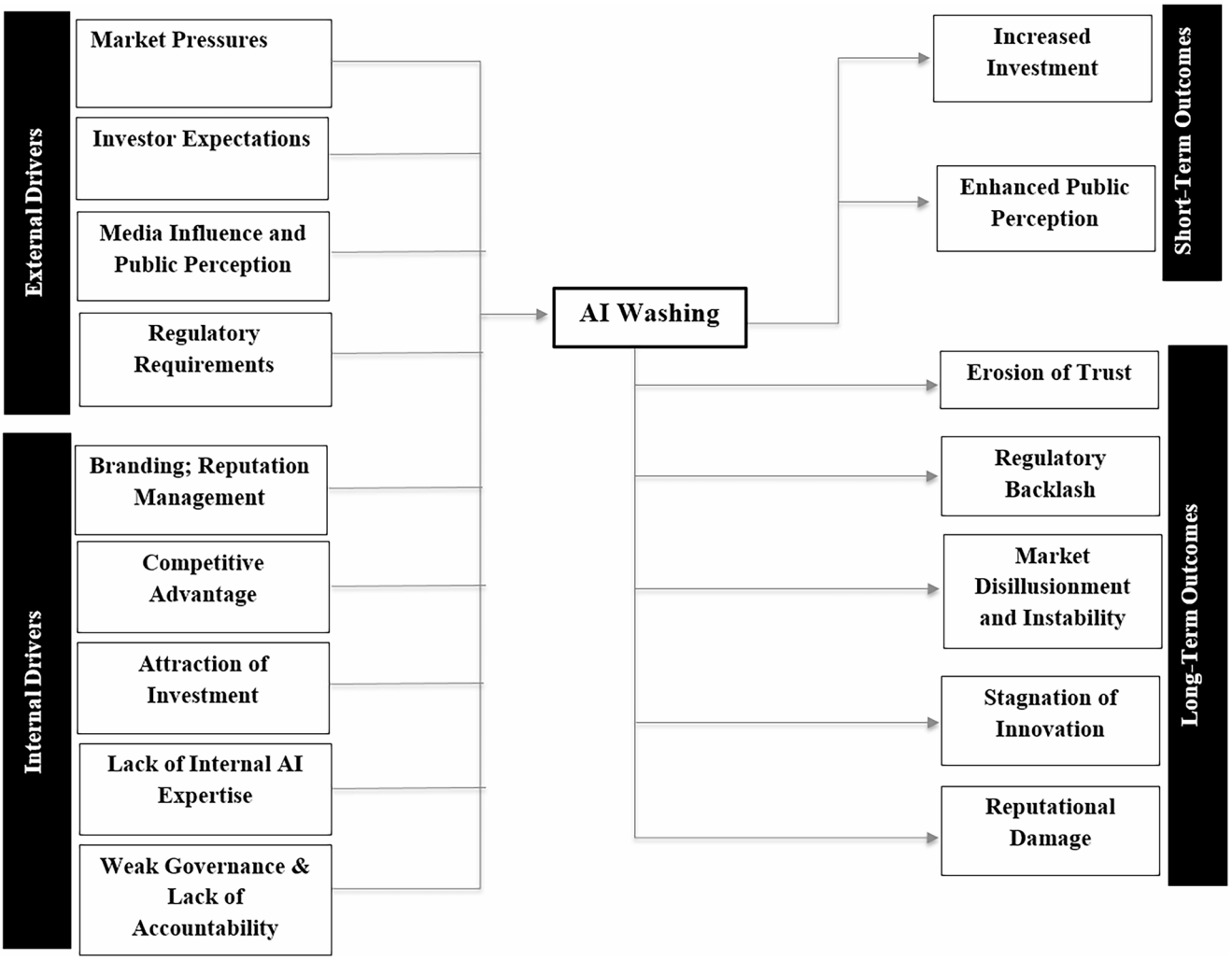


Fig. 1 Conceptual framework of AI washing

responsible AI is becoming subject to regulation, companies may feel compelled to overstate their compliance efforts to avoid potential legal or reputational consequences.

Internal drivers

Internally, several organizational factors contribute to the prevalence of AI washing. One of the key drivers is branding and reputation management. Companies may intentionally exaggerate their use of AI to build or sustain a public image that aligns with innovation and technological leadership. This practice is often linked to efforts aimed at gaining a competitive advantage. By portraying themselves as more advanced in AI than they genuinely are, firms attempt to differentiate themselves in crowded markets, capture a larger market share, or assert dominance within their industry.

The pursuit of investment also plays a critical role. Companies may highlight or exaggerate their AI capabilities to appeal to investors who view AI as a strategic area of growth, particularly in sectors where emerging technologies are perceived as valuable assets. Another internal factor is the lack of genuine AI expertise within the organization. Employees and decision-makers may not fully understand what qualifies as artificial intelligence, leading them to mislabel or overestimate the intelligence of existing systems.

Finally, weak internal governance and the absence of precise accountability mechanisms can create an environment where misleading claims go unchecked. Without proper oversight, ethical guidelines, or compliance processes in place, it becomes easier for companies to make exaggerated or false assertions about their AI capabilities without facing immediate consequences.

Table 3 Theoretical basis for our conceptual framework

Component	Description	Theoretical basis	References
External Drivers	Market pressures, investor expectations, media influence, and regulatory requirements push companies to exaggerate their AI capabilities.	Signaling Theory: Companies send misleading signals to the market, positioning themselves as AI-driven to attract investment and a competitive advantage. This aligns with the idea that firms use AI-related claims to enhance their perceived value. Stakeholder Theory: Firms respond to pressures from investors, customers, and regulators, who expect them to make technological advancements. Even if companies lack substantive AI capabilities, they engage in AI washing to meet these expectations and maintain stakeholder confidence. Legitimacy Theory: Organizations conform to prevailing industry trends, exaggerating AI adoption to maintain legitimacy in the eyes of regulators and the public. Firms may adopt superficial AI initiatives to align with institutional norms, avoiding scrutiny and remaining competitive. Ethics Washing Theory: Companies involved in ethics washing employ misleading language, visuals, omissions, symbolic actions, and lobbying efforts to attract investment and align with market and competitive pressures.	Attard-Frost et al. (2023); Bietti (2020); Connelly et al. (2011); Hagendorff (2020); Seele and Schultz (2022); Suchman (1995); Seele and Schultz (2022); Wagner (2018)
Internal Drivers	Reputation management and competitive advantage drive firms to position themselves as AI leaders, even if their AI capabilities are limited.	Signaling Theory: Companies exaggerate AI capabilities to create an impression of technological sophistication, ensuring that their brand is perceived as forward-thinking and competitive. This signaling helps firms gain customer trust, market valuation, and investor interest. Stakeholder Theory: Firms cater to diverse stakeholders by showcasing AI integration, even when actual capabilities are minimal. The need to satisfy market expectations prompts firms to misrepresent their AI involvement. Legitimacy Theory: Companies shape their public perception by strategically aligning with industry-wide trends in AI adoption. By claiming AI capabilities, firms aim to gain legitimacy among peers, regulatory bodies, and the public, even when these claims are unfounded. Ethics Washing Theory: A lack of accountability, transparency, and governance enables firms to engage in ethics washing, which subsequently extends to AI washing. Unethical organizations are increasingly using AI washing as a routine way to conceal their ethical failings.	Attard-Frost et al. (2023); Connelly et al. (2011); Floridi (2019); Green (2021); Thomas and Lamm (2012)
Mechanisms of AI Washing	Companies engage in symbolic AI washing, attention deflection, or deceptive manipulation, drawing from strategies observed in greenwashing.	Legitimacy Theory: Firms engage in symbolic compliance by making AI-related claims without substantial technological backing. This aligns with 'decoupling,' where organizations present an AI-centric image while operationally maintaining traditional processes. Stakeholder Theory: Firms selectively disclose AI initiatives to appeal to key stakeholders and maintain credibility among investors, consumers, and industry partners. They may highlight minor AI-driven features while neglecting broader AI implementation gaps. Signaling Theory: Despite limited real AI capabilities, companies use attention-grabbing AI narratives to position themselves as industry leaders, influencing investment decisions and shaping consumer perceptions. Ethics Washing Theory: Unethical practices serve as tools for AI washing, enabling the achievement of specific goals. Companies employ these tactics to seem responsible in AI development while concealing real ethical problems, making ethics washing a widespread strategy to evade accountability and preserve a favorable reputation.	Friedman and Miles (2002); Green (2021); Mittelstadt et al. (2016); Morley et al. (2021); Pasquale, 2020a, b; Schultz and Seele, (2023); Thomas and Lamm (2012)
Short-Term Outcomes	AI washing can lead to increased investment and a more positive public perception.	Signaling Theory: Firms leverage AI narratives to gain short-term investor confidence and favorable media coverage, influencing stock prices and market positioning. The perception of AI leadership attracts funding, partnerships, and customer engagement. Legitimacy Theory: Companies gain immediate credibility by aligning themselves with cutting-edge technology trends. By projecting an AI-driven identity, firms mitigate scrutiny and attract favorable regulatory treatment, ensuring market relevance. Ethics Washing Theory: The practice of AI washing, combined with ethics washing, can create a false impression of responsibility in AI development. This helps businesses conceal ethical issues while avoiding accountability and maintaining a positive public image, all while incurring lower costs and assuming less responsibility.	Attard-Frost et al. (2023); Bietti (2020); Connelly et al. (2011); Floridi (2019); Seele and Schultz (2022); Thomas and Lamm (2012);

Table 3 (continued)

Component	Description	Theoretical basis	References
Long-Term Outcomes	Over time, AI washing may lead to erosion of trust, regulatory backlash, market disillusionment, stagnation of innovation, and reputational damage.	Stakeholder Theory: Over time, as investors, regulators, and consumers recognize AI misrepresentation, firms face backlash. Customers lose trust, investors withdraw, and regulatory bodies introduce stricter oversight, penalizing deceptive claims made by AI. Legitimacy Theory: When the disparity between AI claims and actual capabilities becomes evident, firms suffer reputational damage. Loss of legitimacy leads to decreased stakeholder confidence, a reduced competitive advantage, and potential legal repercussions. Signaling Theory: The long-term consequence of misleading AI signaling is market skepticism. Firms that fail to deliver on AI promises face difficulty regaining trust, leading to innovation stagnation and a loss of investor interest. Ethics Washing Theory: Ethics Washing Theory explains that AI washing, as a form of ethics washing, damages a firm's ethical culture and reputation. It erodes long-term trust and accountability by enabling companies to conceal ethical issues, thereby risking significant harm to their credibility and relationships with stakeholders.	Attard-Frost et al. (2023); Green (2021); Floridi (2019); Friedman and Miles (2002); Morley et al. (2021); Schultz and Seele (2023)

Outcomes of AI washing

The consequences of AI washing can yield short-term benefits, such as increased visibility and investor interest, but often result in long-term drawbacks, including reputational damage and loss of trust. These effects impact various stakeholders, including customers, employees, investors, and industry partners.

Short-term outcomes

In the short term, AI washing can lead to a surge in investment, as venture capitalists and institutional investors are often drawn to companies that appear to be at the forefront of AI innovation. This perception of technological leadership can also enhance a company's public image. By successfully presenting themselves as AI-driven, firms may improve their brand reputation and gain recognition as forward-thinking and innovative players in their industry.

Long-term consequences

Over time, AI washing can lead to significant and lasting consequences. One major impact is the erosion of trust. When stakeholders such as investors, customers, or regulators uncover the gap between a company's claims and its actual capabilities, confidence in the organization can diminish, resulting in reputational damage and a decline in customer loyalty. This loss of credibility is often difficult to reverse and can affect future business prospects.

Another serious consequence is regulatory backlash. If the AI washing involves misrepresentations that breach ethical guidelines or regulatory standards, companies may face intensified scrutiny, legal challenges, or financial penalties from authorities. Beyond legal risks, AI washing can contribute to broader market disillusionment. As exaggerated claims become more prevalent, skepticism about AI solutions may increase across industries, potentially slowing the adoption of genuine innovations. This disillusionment can also spark speculative bubbles, inflating valuations based on misleading information and eventually leading to investor losses and instability, much like what has been observed in cases of greenwashing.

Internally, companies that prioritize appearances over substance may fail to invest in genuine AI capabilities. This short-sighted approach can result in a stagnation of innovation, preventing firms from keeping pace with technological progress and undermining their long-term competitiveness. Finally, reputational harm often lingers. A company exposed for AI washing may be perceived as dishonest or opportunistic, which can damage its ability to attract future customers, secure investments, or recruit top talent. The reputational cost can mirror the long-lasting damage experienced by firms implicated in greenwashing controversies.

Mechanisms of AI washing

Firms engaging in AI washing employ various interrelated strategies to create the illusion of technological sophistication and ethical responsibility. These mechanisms reflect broader organizational behaviors centered on legitimacy seeking, stakeholder influence, reputational signaling, and ethical posturing. Rather than being isolated or accidental, these practices are systematic and strategic, crafted to shape perceptions while evading the significant commitments required for authentic AI integration or ethical reform.

From a legitimacy theory perspective, firms often engage in symbolic compliance by making AI-related claims that lack substantive technological foundations. This behavior

aligns with decoupling, where organizations maintain a public image of AI transformation through branding, executive communication, or sustainability reports, while their internal processes remain largely unaltered by advanced AI technologies (Thomas & Lamm, 2012). These actions help companies conform to societal expectations of innovation and digital transformation, thereby securing legitimacy in the eyes of regulators, industry observers, and the broader public, without incurring the operational risks or resource burdens associated with genuine change.

Stakeholder theory further explains how firms manage trust and legitimacy through selective disclosure. Organizations curate narratives to resonate with different stakeholder groups, highlighting minor AI-driven features such as chatbots or personalized recommendations while omitting broader implementation gaps and unresolved ethical concerns (Friedman & Miles, 2002). This selective transparency enables firms to appeal to investors eager for innovation, consumers drawn to personalization, and policymakers demanding accountability, even when AI engagement remains limited or experimental.

The strategic use of language plays a critical role. Drawing on signaling theory, firms utilize terms like "AI-powered," "machine learning-enabled," or "autonomous" to construct narratives that convey innovation leadership and future-readiness (Connelly et al., 2011; Pasquale, 2020a, b). These signals aim to attract media attention, influence investor behavior, and differentiate products in the marketplace. The effectiveness of such messaging is heightened by the information asymmetry between firms and their audiences, many of whom lack the technical expertise to verify the accuracy or depth of these AI claims. As a result, firms can capitalize on AI's cultural cachet without delivering corresponding technological value.

In parallel, AI washing often draws from practices associated with ethics washing. Here, firms signal their commitment to responsible AI through performative gestures, such as publishing ethical principles, forming advisory boards, or affiliating with high-profile industry initiatives, without implementing robust ethical oversight or accountability mechanisms (Schultz & Seele, 2023; Seele & Schultz, 2022). These symbolic actions serve as a moral smokescreen, allowing organizations to claim alignment with societal values while continuing practices that may involve algorithmic bias, data exploitation, or a lack of transparency. The appearance of ethical engagement becomes a reputational asset, while the underlying issues remain unaddressed.

Together, these mechanisms reveal that AI washing is a deliberate and layered organizational response to external pressures—one that allows firms to project technological and ethical legitimacy while sidestepping the structural, financial, and moral demands of genuine AI transformation.

Typology of AI washing

AI washing is not a one-size-fits-all practice; it varies depending on the exaggeration, intent, and complexity of misrepresenting AI. Building upon the frameworks provided in the works on greenwashing (Sinao et al., 2017; Marquis & Toffel, 2011; Guo et al., 2018; Du et al., 2018), we propose a typology for "AI Washing." Just as greenwashing involves the disparity between symbolic actions and genuine sustainability, AI washing refers to companies' deceptive portrayal of AI capabilities.

From Siano et al.'s (2017) exploration of deceptive manipulation within greenwashing, we draw parallels to AI washing, where organizations present themselves as leveraging AI when they are not, or only minimally doing so. Just as greenwashing can be divided into decoupling (claiming sustainability without actual implementation) and attention deflection (misleading the public with selective information), AI washing can be categorized into the following types (see Table 4 for a summary of the categories).

The first type is symbolic AI washing, where companies exaggerated or made false claims about the sophistication of their AI tools, similar to the 'empty green claims' seen in greenwashing (Marquis & Toffel, 2011; Guo et al., 2018). This can involve mislabelling basic automation as AI or overstating AI's impact within the organization. Symbolic AI washing primarily concerns creating an image of innovation rather than integrating AI into core business operations. These companies often use vague, non-specific language about AI designed to generate media attention or project a futuristic image, without substantively implementing

Table 4 Typology of AI washing

Type of AI washing	Strategic intent	Key tactics	Example behavior
1. Symbolic AI Washing	Image management, media hype, brand innovation	- Vague claims - Buzzword usage - Conference sponsorship - Overstating minor AI usage	Marketing generic "AI insights" without straightforward implementation
2. Attention Deflection AI Washing	Divert scrutiny; investor/market appeasement	- Selective disclosure - Highlighting narrow AI use - Ignoring broader non-AI reliance	Showcasing one AI-driven tool while the bulk of operations remain manual
3. Deceptive Manipulation AI Washing	Regulatory appeasement, ethics-washing, legal shield	- Falsified transparency - Misleading ethics claims - Suppressing real data practices	Claiming AI ethics compliance without fundamental guidelines, hiding model/data use practices

the technology. For instance, businesses may employ AI-related buzzwords in their marketing or sponsor AI conferences to align themselves with innovation, even if the actual AI deployment is minimal. This tactic often aims to shape public perception and enhance the brand rather than reflect genuine technological advancement. An example might be a retail brand touting 'AI-driven insights' without explaining how AI is meaningfully integrated into its operations.

The second type is attention deflection AI washing, which involves highlighting minor applications of AI to divert attention from the company's significant reliance on traditional processes. This strategy mirrors the 'halo effect' or 'selective disclosure' observed in greenwashing (Du et al., 2018). In this case, companies strategically concentrate on the limited areas where AI is used, creating a misleading narrative of widespread AI-driven innovation. At the same time, their broader operations remain dependent on outdated or manual systems. This form of AI washing often targets investors or regulators, intending to diminish scrutiny or inflate the company's perceived innovation. Examples include labelling simple automation or decision-tree algorithms as sophisticated AI or overstating the potential of recommendation engines and predictive analytics. Such actions create the illusion of technological advancement without business transformation.

The third type is deceptive manipulation AI washing, which reflects a more ethically questionable practice, where companies make substantial organizational or technical changes to appear as if they are using AI extensively, even when this is not the case. This approach often pushes the boundaries of ethical and legal standards in AI claims. Companies may claim to follow AI ethics guidelines or transparency measures in this form, but these claims are often misleading or unsubstantiated. Deceptive manipulation AI washing is usually driven by regulatory requirements or industry expectations and is aimed at gaining the trust of regulators or industry watchdogs (Floridi et al., 2018a, b). A company may promote ethical AI practices without supporting policies or hiding key information about AI models and data usage while marketing transparency initiatives.

AI washing future research questions

Despite AI's growing prominence in business, the literature on AI washing—where companies exaggerate or misrepresent their use of AI—remains in its infancy. This phenomenon has significant implications for consumer behavior and marketing strategy, requiring a more thorough investigation.

AI washing extends beyond ethical concerns, influencing competitive dynamics and strategic decision-making. Its significance is evident not only in literature but also in business practice, where transparency and authenticity in AI-related claims are critical. However, the nuances of AI washing, its underlying motivations, and its short—and long-term outcomes have not been adequately explored.

To address these knowledge gaps, we propose three categories of research questions. Our goal is to delve deeper into how consumers perceive AI washing, the impact of marketing communications on these perceptions, and the role of regulations in mitigating deceptive AI claims. Moreover, we aim to study the strategic aspects that affect AI washing practices and their influence on competitive advantage and brand reputation. Finally, we focus on how scholarly research can support public policy needed to address AI washing and create rigorous methods to uncover various AI washing activities.

Drawing upon our comprehensive literature review and relevant theoretical frameworks, these research questions offer potential avenues for future academic investigation and practical application, contributing to a more nuanced understanding of the AI washing phenomenon. The research questions proposed are summarized in Table 5. By exploring these research questions, additional contributions can be provided for a more comprehensive understanding of AI washing and its implications for consumers, businesses, and society. Although we provide a list of research questions, we believe that additional questions can be developed based on this study.

Discussion

The growing prevalence of artificial intelligence in marketing has intensified both innovation and misrepresentation. Firms increasingly adopt AI narratives not only as strategic tools but also as symbolic assets that communicate progressiveness, efficiency, and technological sophistication. Against this backdrop, this study introduces and conceptualizes AI washing as an emergent ethical issue in marketing communication. Much like greenwashing in environmental contexts (Delmas & Burbano, 2011) and ethics washing in corporate social responsibility (Lyon & Montgomery, 2015a, b), AI washing reflects a misalignment between claimed and actual practices, often for reputational or competitive gain. However, despite its rising prevalence in business rhetoric, AI washing remains theoretically underdeveloped. This paper addresses that gap by offering both a typology and an ethics-grounded conceptual framework.

The typology advances the literature on deceptive marketing by categorizing AI washing based on the degree of misrepresentation, its intentionality, and its target audience. Prior research has shown how firms use symbolic language to project legitimacy and conform to institutional expectations (Suchman, 1995; Elsbach & Sutton, 1992). Building

Table 5 AI washing research questions

Consumer Behavior and AI washing
What are the ethical concerns of using AI washing in marketing and communication?
How do consumers perceive the implications of AI washing, and does this impact their attitudes and purchasing decisions?
Are there cultural or demographic differences in consumer perceptions and attitudes toward AI washing?
How do consumers evaluate AI-driven marketing or communication claims, and what factors influence their trust? What is the role of AI washing in such influence?
What are consumers' expectations regarding using AI in marketing and communication?
How can AI washing research be used to enlighten consumer education? How can consumers be empowered to discover AI washing?
Marketing Strategy and AI Washing
How do brands justify using AI washing in their marketing and communication strategies?
What is the relationship between AI washing, brand reputation, customer satisfaction, and financial performance?
How do brands that engage in AI washing compare to those that do not in terms of brand reputation, customer satisfaction, and financial performance?
How can reputation failure caused by AI washing be managed or recovered?
What are the potential long-term consequences of AI washing for brands, including legal, reputational, and financial risks? What are the mechanisms of these consequences?
Public Policy and AI Washing
What are the existing regulatory frameworks and guidelines for AI use in marketing that can be used to address AI washing? Are these frameworks and guidelines sufficient? What new frameworks and guidelines are needed? How to create effective frameworks and guidelines?
How can policymakers and industry bodies collaborate to address AI washing? How can scholarly research help with that? How can AI washing research findings be used to inform policy decisions?
How to uncover AI washing? What are the technological limitations and challenges in detecting and measuring AI washing? How can researchers develop methodologies to identify and quantify AI washing in marketing materials?

on these insights, our typology positions AI washing as a form of symbolic communication that seeks legitimacy through technological signaling rather than substantive innovation. In particular, we extend the application of signaling theory (Spence, 1973) to the AI context, where ambiguous or unverifiable AI claims are used as strategic signals to influence investor perceptions, customer interest, and stakeholder evaluations.

Moreover, by grounding AI washing in the broader framework of ethics washing (Floridi, 2022) and integrating the Hunt–Vitell model of ethical decision-making (Hunt & Vitell, 1986; Vitell et al., 2001), we provide a normative lens to understand how and why firms engage in ethically questionable AI narratives. The Hunt–Vitell model is especially useful in this context because it combines deontological reasoning (based on rules and duties) and teleological reasoning (based on outcomes), offering a dual-process view of how marketers evaluate the morality of their decisions. In applying this model, we argue that many AI washing practices arise from perceived instrumental benefits—such as market attention, funding, or differentiation—while firms simultaneously rationalize these practices as industry norms or communication shortcuts.

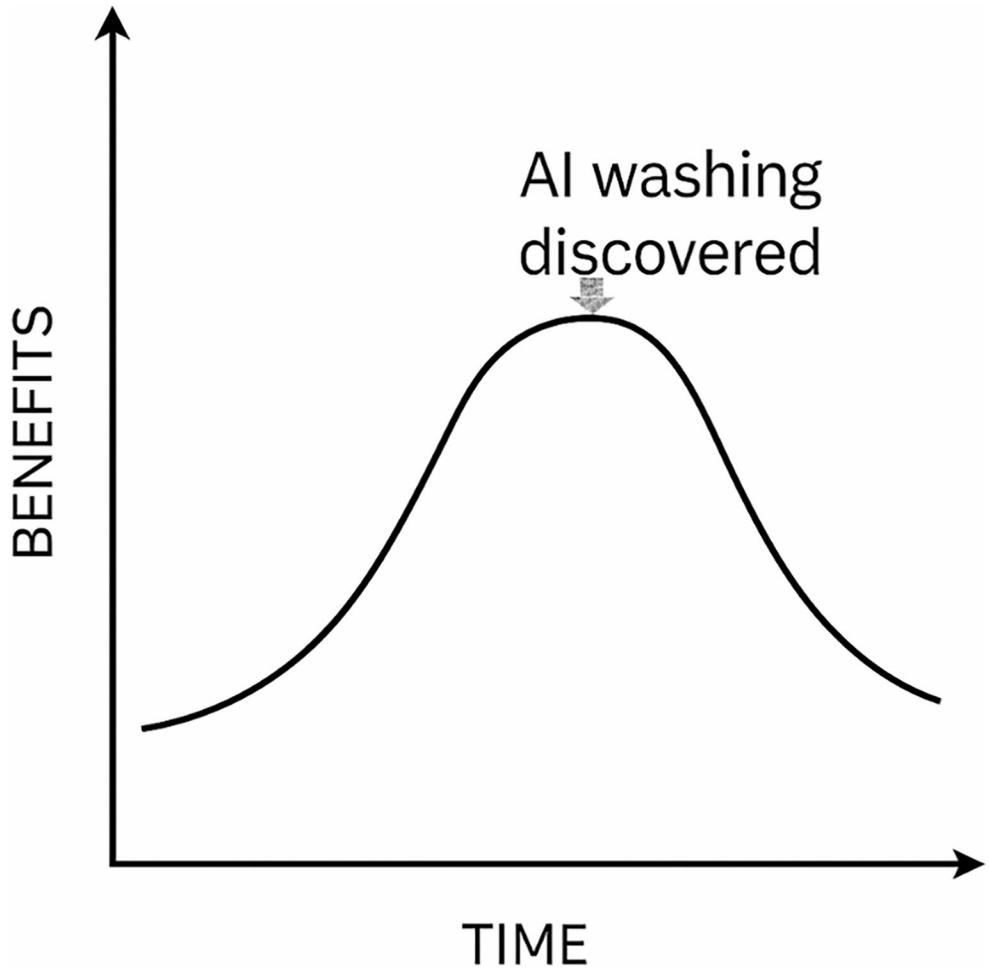
This dual framing links our work to a broader conversation about moral flexibility in marketing ethics (Ferrell et al., 2024), where firms often navigate ethical gray zones depending on competitive pressures, stakeholder expectations, and internal norms. The study also contributes to calls in marketing literature for a more nuanced examination of "ethics in action" (Laczniak & Murphy, 2019), especially in contexts where rapid technological change outpaces the development of ethical standards or regulatory guidance.

One particularly relevant aspect of AI washing is its impact on trust, which is central to marketing relationships. While AI is frequently presented as a tool for enhancing trust—through automation, personalization, and predictive capabilities—false or inflated claims can undermine that very trust. This echoes the paradox identified in digital trust literature (Sundararajan, 2019) where technological sophistication is both a source of competitive advantage and a potential threat to consumer confidence. Our findings suggest that when trust is based on unverifiable or opaque AI claims, firms risk long-term reputational damage if these claims are later revealed to be exaggerated or misleading.

Importantly, the study also draws attention to the B2B dimension of AI washing, which has been largely overlooked in previous research. While much of the ethical marketing discourse focuses on protecting consumers, some of the most consequential AI misrepresentations target investors, regulators, and strategic partners. These audiences are often less concerned with product functionality and more attuned to narratives of innovation and scalability. The use of buzzwords like "AI-powered" or "machine learning-enabled" functions as a signaling mechanism in high-stakes corporate communications, especially during fundraising rounds, acquisition discussions, or public disclosures. This aligns with research on narrative legitimacy (Navis & Glynn, 2011), which shows that organizational storytelling shapes how firms are evaluated by external audiences.

Beyond these contributions, our research highlights that AI washing is not just a communication issue but also a strategic and ethical one. Firms that rely heavily on AI narratives without delivering corresponding value may experience short-term gains but are likely to face long-term consequences. These can include misaligned internal resource allocation, declining consumer trust, greater regulatory scrutiny, and brand vulnerability. The tension between signaling and substance reinforces earlier critiques

Fig. 2 AI washing effect



of technology hype cycles (Wang & Ramiller, 2009), where the pressure to appear cutting-edge leads to premature or misleading technological adoption claims.

Conclusion

AI washing involves exaggerating or falsely claiming the use of artificial intelligence in products or services to attract customers and investors. This deceptive practice misleads buyers by overstating capabilities, causing unrealistic expectations and potential financial loss. It also blurs the line between genuine AI innovations and superficial claims, damaging trust in companies and the broader AI industry. The benefits that firms can receive from AI Washing are short-term, and once discovered, these benefits will vanish (See Fig. 2), similar to what happened with Amazon's Just Walk Out technology.

The phenomenon of AI washing presents a significant challenge in the rapidly evolving AI landscape. Firms can gain short-term benefits by intentionally exaggerating or misrepresenting their AI capabilities, such as increased investment and enhanced public perception. However, the long-term consequences of AI washing can be detrimental, including the erosion of trust, regulatory backlash, market disillusionment, and stagnation of innovation.

Our study advances the conceptual understanding of AI washing in marketing through three key contributions. First, it offers a clear definition and structured typology that distinguishes forms of AI washing across B2B and consumer markets. Second, it grounds the phenomenon in marketing ethics by integrating the broader concept of ethics washing, explaining how AI washing emerges and why it matters. Third, it provides actionable strategies for firms, regulators, and platforms to mitigate AI washing and promote responsible AI communication.

Our study also enhances the understanding of AI washing by extending existing theories of ethical misrepresentation and marketing practices into the realm of AI. By developing a typology of AI washing, we establish a foundation for future studies to explore the nuances of this phenomenon further. Our framework opens new avenues for analyzing how AI is marketed and the ethical considerations that arise when companies misrepresent their technological

capabilities. Moreover, this work contributes to the growing body of literature on ethical AI by linking it to marketing ethics and consumer trust, providing a basis for empirical studies.

From a practical perspective, this paper offers actionable insights for marketers, business leaders, and regulators. Marketers must be aware of the risks associated with AI washing, particularly the potential erosion of consumer trust and the long-term reputational damage that can result from misrepresenting AI capabilities. Thus, transparency in AI-related marketing claims is essential to maintaining credibility and avoiding the pitfalls of overpromising technological features. We recommend that companies adopt stringent internal auditing processes for AI claims and collaborate closely with their technical teams to represent AI functionalities accurately.

To effectively address AI washing, coordinated action is essential from investors, consumers, regulators, and companies. These roles are best understood through the theoretical lenses discussed in this paper: signaling theory, stakeholder theory, legitimacy theory, and ethics washing theory.

From the perspective of signaling theory, investors should require transparent, verifiable information about companies' AI capabilities. Since firms often signal innovation to attract capital, reinforcing expectations for evidence-based AI claims helps align market perceptions with actual technological adoption. Investors act as gatekeepers in filtering out superficial or misleading AI narratives.

Stakeholder theory positions consumers as empowered participants in the broader accountability network. Educating consumers through AI and digital literacy initiatives strengthens their capacity to critically assess AI-marketed products, reducing susceptibility to vague or inflated claims. Informed consumers apply pressure on firms to offer authentic value, thus reinforcing stakeholder-driven standards for responsible innovation.

In line with legitimacy theory, regulators must define and enforce clear guidelines around AI claims to protect institutional credibility and public trust. Regulatory frameworks should require precise definitions of AI functionalities, mandate disclosure standards, and penalize deceptive marketing practices. These measures not only curb misinformation but also help maintain the legitimacy of the technological ecosystem.

Ethics washing theory highlights the propensity of organizations to adopt superficial ethical narratives in response to external scrutiny. This theoretical lens emphasizes the importance of embedding ethical principles into governance structures and operational practices, rather than relying on symbolic declarations or unverified commitments. When companies fail to implement substantive ethical measures, they risk long-term reputational harm, diminished stakeholder trust, and broader disillusionment with AI technologies across the industry.

Ultimately, the mitigation of AI washing depends on the collective action of all stakeholders. Each actor must fulfill their role not simply as a matter of preference or compliance, but as a response to the theoretical imperatives of trust signaling, stakeholder responsibility, institutional legitimacy, and ethical integrity. This multi-layered approach ensures that AI is integrated meaningfully and transparently into business practices, thereby fostering a more credible and sustainable AI ecosystem.

While this paper lays the conceptual foundation for understanding AI washing, future research is needed to empirically validate the proposed typology and examine the impact of AI washing on various stakeholders. Researchers could investigate how different forms of AI misrepresentation affect consumer trust, purchasing behavior, and brand loyalty. Another important avenue for future work is the development of frameworks to detect AI washing practices using machine learning or natural language processing. Our study also highlights the need for cross-disciplinary collaborations between AI developers, ethicists, and marketers to ensure more ethical marketing practices.

However, this conceptual paper has limitations. The lack of empirical testing means that the typology requires validation in real-world settings. Furthermore, although we have drawn from the existing washing literature, AI washing is an emerging phenomenon, and its impacts may evolve as AI technologies and public awareness continue to advance. Future research could address these limitations by conducting longitudinal studies to explore how AI washing practices evolve and by analyzing the effectiveness of potential regulatory responses.

Author contributions Equal contributions

Funding Not applicable.

Data availability Not applicable

Declarations

Ethical approval Not applicable.

Competing interest Not applicable.

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